STA2002: Probability and Statistics II
One-way ANOVA

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Introduction

- In this lecture we will one-way analysis-of-variance (ANOVA). Frequently, experimenters want to compare more than two treatments, e.g., yields of several different corn hybrids, results due to three or more teaching techniques, etc.
  Sometimes the different treatment distributions of the resulting observations are due to changing the level of a certain factor (e.g., different doses of a given drug). Thus, the consideration of the equality of the different means of the various distributions comes under the analysis of a one-factor experiment.
- **Suggested reading**: Chapter 9.3 of the textbook.

Introduction

- Previously, we have discussed how to compare the means of two normal distributions.
- More generally, we now consider *m* normal distributions with **unknown** means $\mu_1, \mu_2, \ldots, \mu_m$ and an **unknown**, but common, variance σ^2.
- Usually, we care about a test of the equality of the *m* means, namely,

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_m = \mu, \tag{1}$$

with μ **unspecified**, against all possible alternative hypotheses H_1.

Introduction

- In order to test this null hypothesis H_0, we need to take **independent random samples** from these distributions, respectively.
- Let $X_{i1}, X_{i2}, \ldots, X_{in_i}$ represent a random sample of size n_i from the normal distribution $N(\mu_i, \sigma^2)$, $i = 1, 2, \ldots, m$.
- Usually, we can summarize the data points in a table, see the attached table on the next page.

Format of Random Samples

a data structure — data matrix

				Means
X_{11}	X_{12}	$\cdots$	X_{1n_1}	$\overline{X}_{1\cdot}$
X_{21}	X_{22}	$\cdots$	X_{2n_2}	$\overline{X}_{2\cdot}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
X_{m1}	X_{m2}	$\cdots$	X_{mn_m}	$\overline{X}_{m\cdot}$

Introduction

- **Sample means** are defined as,

$$\overline{X}_{i\cdot} := \sum_{j=1}^{n_i} X_{ij}/n_i, \ i = 1, 2, \ldots, m.$$

- **Grand mean** is defined as,

$$\overline{X}_{\cdot\cdot} := \sum_{i=1}^{m} \sum_{j=1}^{n_i} X_{ij}/(n_1 + n_2 + \cdots + n_m).$$

- The total sample size of all random samples is,

$$n = n_1 + n_2 + \cdots + n_m.$$

To Construct Test Statistic

- Question: How to construct a test statistic and propose the relative natural form of the critical region, for H_0 proposed in Expression (1).

- Later, we manipulate the following quantity,

$$\text{SS(TO)} := \sum_{i=1}^{m} \sum_{j=1}^{n_i} (X_{ij} - \overline{X}_{\cdot\cdot})^2$$

  which is called **the total sum of squares**.

- SS(TO) would be partitioned into two parts, one is called **the error sum of squares**, and the other is called **the between-treatment sum of squares**.

To Construct Test Statistic

- Namely,

$$\begin{aligned} \text{SS(TO)} &= \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij}-\overline{X}_{i\cdot})^2 + \sum_{i=1}^{m} n_i(\overline{X}_{i\cdot}-\overline{X}_{\cdot\cdot})^2 \\ &= \text{SS(E)} + \text{SS(T)}, \end{aligned}$$

where

$$\text{SS(E)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij}-\overline{X}_{i\cdot})^2,$$

$$\text{SS(T)} = \sum_{i=1}^{m} n_i(\overline{X}_{i\cdot}-\overline{X}_{\cdot\cdot})^2.$$

Sum of Squares

$$\mathrm{SS(TO)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{..})^2, \text{ the total sum of squares;}$$

$$\mathrm{SS(E)} = \sum_{i=1}^{m}\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i.})^2, \quad \text{the sum of squares within treatments, groups, or classes, often called the error sum of squares;}$$

$$\mathrm{SS(T)} = \sum_{i=1}^{m} n_i\,(\overline{X}_{i.} - \overline{X}_{..})^2, \quad \text{the sum of squares among the different treatments, groups, or classes, often called the between-treatment sum of squares.}$$

To Construct Test Statistic

- Under $H_0: \mu_1 = \mu_2 = \cdots = \mu_m = \mu$, we may regard X_{ij}, $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n_i$, as one random sample of size $n = n_1 + n_2 + \cdots + n_m$ from the normal distribution $N(\mu, \sigma^2)$.
- Under H_0,

$$\frac{\text{SS(TO)}}{(n-1)}$$

is actually the sample variance of random sample with size n, which is an unbiased estimator for σ^2.

- Recall Student's Theorem which is reviewed in Slides 1.

To Construct Test Statistic

- Under H_0,

$$\frac{\text{SS(TO)}}{\sigma^2}$$

follows a $\chi^2(n-1)$ distribution, with

$$E\left[\frac{\text{SS(TO)}}{\sigma^2}\right] = n-1,$$

and

$$E\left[\frac{\text{SS(TO)}}{n-1}\right] = \sigma^2.$$

To Construct Test Statistic

- Similarly, construct statistic W_i as follows,

$$W_i := \frac{\sum_{j=1}^{n_i}(X_{ij} - \overline{X}_{i\cdot})^2}{n_i - 1}, \quad i = 1, 2, \ldots, m,$$

  which is the sample variance in the i'th distribution.

- We have that,

$$\frac{(n_i - 1)W_i}{\sigma^2}$$

  follows a $\chi^2(n_i - 1)$ distribution, with

$$E\left[\frac{(n_i - 1)W_i}{\sigma^2}\right] = n_i - 1, \quad E[W_i] = \sigma^2.$$

To Construct Test Statistic

- Do more observations, and we have that,

$$\text{SS(E)} = \sum_{i=1}^{m} (n_i - 1) W_i$$

where $W_1, w_2, \ldots, W_m$ are independent.

- Thus, we have that,

$$\frac{\text{SS(E)}}{\sigma^2}$$

follows a chi-square distribution with degrees of freedom $(n_1 - 1) + (n_2 - 1) + \cdots + (n_m - 1) = n - m$.

To Construct Test Statistic

- The expectation of $\frac{\text{SS(E)}}{\sigma^2}$ is,

$$E\left[\frac{\text{SS(E)}}{\sigma^2}\right] = n - m$$

and then,

$$E\left[\frac{\text{SS(E)}}{n-m}\right] = \sigma^2.$$

- **Remark.** Recall that for chi-square distribution, the mean equals the d.f.

To Construct Test Statistic

- We have already prepared the following partition,

$$\frac{\text{SS(TO)}}{\sigma^2} = \frac{\text{SS(E)}}{\sigma^2} + \frac{\text{SS(T)}}{\sigma^2}. \tag{2}$$

- Under H_0, $\frac{\text{SS(TO)}}{\sigma^2}$ follows a chi-square distribution with d.f. $n-1$.
- No matter H_0 is true or not, $\frac{\text{SS(E)}}{\sigma^2}$ follows a chi-square distribution with d.f. $n-m$.
- Moreover, SS(T) ≥ 0, and it can be proved that SS(E) and SS(T) are independent. → under H_0.

An Useful Theorem

Theorem 1 (Theorem 9.3.1)

Let $Q = Q_1 + Q_2 + \cdots + Q_k$, where $Q, Q_1, ..., Q_k$ are $k + 1$ real quadratic forms in n mutually independent random variables normally distributed with the same variance σ^2.

Let $Q/\sigma^2, Q_1/\sigma^2, \ldots, Q_{k-1}/\sigma^2$ have chi-square distributions with $r, r_1, ..., r_{k-1}$ degrees of freedom, respectively. If Q_k is nonnegative, then

(a) $Q_1, \ldots, Q_k$ are mutually independent, and hence,
(b) Q_k/σ^2 has a chi-square distribution with $r - (r_1 + \cdots + r_{k-1}) = r_k$ degrees of freedom.

The proof is not required.

To Construct Test Statistic

- Apply Theorem 1 to Equation (2), we have that, under H_0,

$$\frac{\text{SS(T)}}{\sigma^2}$$

follows a chi-square distribution with degrees of freedom $(n-1)-(n-m)=m-1$.

- Then, we have that, under H_0,

$$E\left[\frac{\text{SS(T)}}{\sigma^2}\right]=m-1,$$

$$E\left[\frac{\text{SS(T)}}{m-1}\right]=\sigma^2.$$

To Construct Test Statistic

- In general, no matter H_0 is true or not,

$$E\left[\frac{\text{SS(T)}}{m-1}\right] = \sigma^2 + \sum_{i=1}^{m} n_i \frac{(\mu_i - \bar{\mu})^2}{m-1},$$

where $\bar{\mu} := \sum_{i=1}^{m} n_i \mu_i / n$.

- Under H_0, $\mu_i = \bar{\mu} = \mu$, $i = 1, 2, \ldots, m$, then

$$\sum_{i=1}^{m} n_i \frac{(\mu_i - \bar{\mu})^2}{m-1} = 0.$$

Test Statistic

- The test statistic is constructed as follows,

$$F := \frac{\mathrm{SS(T)}/(m-1)}{\mathrm{SS(E)}/(n-m)},$$

  which follows an F distribution with $m-1$ and $n-m$ degrees of freedom, under H_0.

- In the case that the means $\mu_1, \mu_2, \ldots, \mu_m$ begin to differ, the value of ratio statistic F tends to become large.

Decision Rule

- **Critical region**,

$$C := \{F \geq F_{\alpha}(m-1, n-m)\},$$

where α is the significance level.

- **p-value**,

$$\textbf{p-value} = P(F(m-1, n-m) > f),$$

where f denotes the realized value of the test statistic F, and $F(m-1, n-m)$ denotes an F distribution with $m-1$ and $n-m$ degrees of freedom.

Example 9.3-2

Example 2

A window that is manufactured for an automobile has five studs for attaching it. A company that manufactures these windows performs “pullout tests" to determine the force needed to pull a stud out of the window.

Let X_i, $i = 1, 2, 3, 4, 5$, equal the force required at position i, and assume that the distribution of X_i is $N(\mu_i, \sigma^2)$.

We shall test the null hypothesis

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5,$$

using seven independent observations at each position.

Example 9.3-2

- Question: At an $\alpha = 0.01$ significance level, whether reject H_0 or retain H_0?
- Data points are summarized as follows,

Table 9.3-5 Pullout test data

	Observations							$\sum_{j=1}^{7} x_{ij}$	$\sum_{j=1}^{7} x_{ij}^2$
x_1:	92	90	87	105	86	83	102	645	59,847
x_2:	100	108	98	110	114	97	94	721	74,609
x_3:	143	149	138	136	139	120	145	970	134,936
x_4:	147	144	160	149	152	131	134	1017	148,367
x_5:	142	155	119	134	133	146	152	981	138,415
Totals								4334	556,174

Example 9.3-2

- Solution.
  The realized value for the test statistic F is

$$f = \frac{16,672.11/4}{2828.86/30} = 44.20,$$

- $F_{0.01}(m-1, n-m) = F_{0.01}(4, 30) = 4.02$. $n = 35$ and $m = 5$.

Table 9.3-6 ANOVA table for pullout tests

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Treatment	16,672.11	4	4, 168.03	44.20
Error	2,828.86	30	94.30	
Total	19,500.97	34		

Example 9.3-2

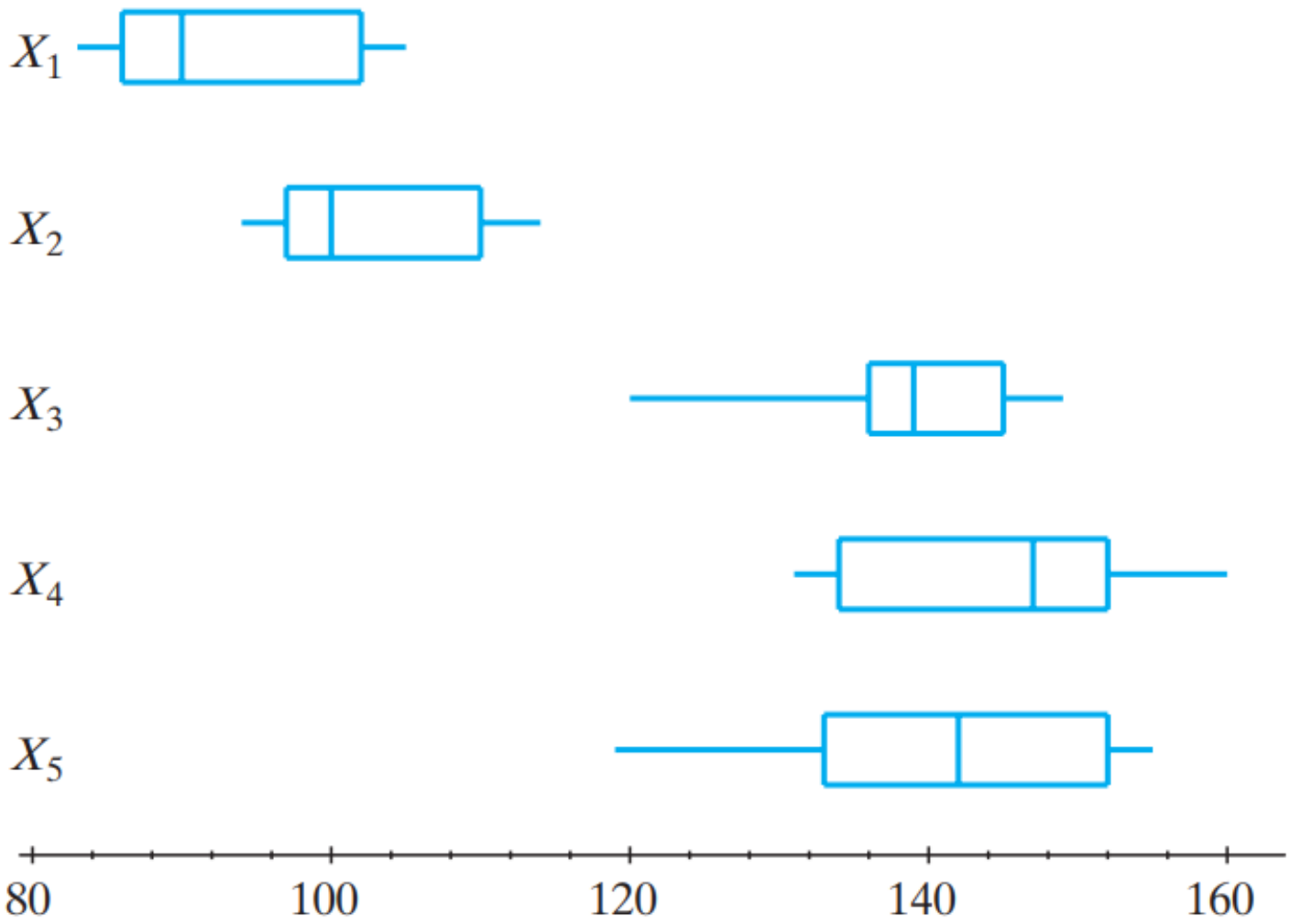


Figure 9.3-1 Box plots for pullout tests

Example 9.3-2

- Decision: Reject H_0.
  With these data, it looks like the forces required to pull out studs in positions 1 and 2 are similar, and those in positions 3, 4, and 5 are quite similar, but different from, positions 1 and 2.

□